

CS331

Sample Midterm Exam

Q1- Propose an entity-relationship (ER) diagram for the following requirements. For each of the entities, feel free to add attributes if needed.

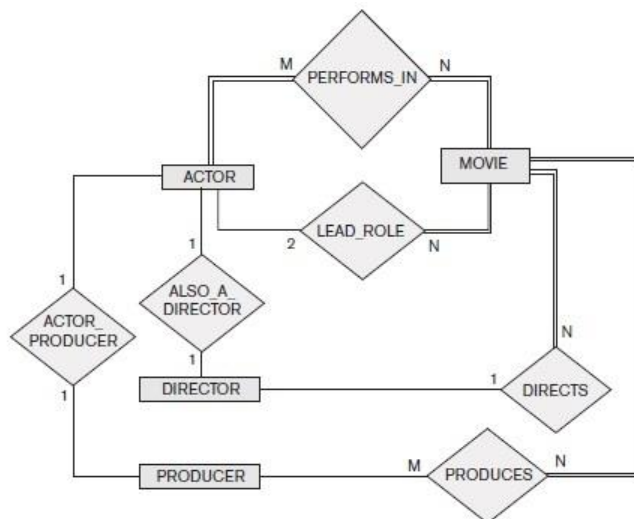
A large organization has several parking lots, which are used by staff.

Each parking lot has a unique name, location, capacity, and number of floors (where appropriate).

Each parking lot has parking spaces, which are uniquely identified using a space number.

Members of staff can request the use of a parking space. Each member of staff has a unique number, name, telephone extension number, and vehicle license number.

Q2- Give a step by step mapping of the following ER diagram to the relational model. Explain your decisions.



Q3- The human resources department of an enterprise maintains a database on the company's employees in order to track their qualifications at various skill levels. All attributes are text data types except DATE_QUALIFIED, which is a date data type. The tables are as follows.

- a-** Identify the primary key(s) and foreign key(s) of each table.
- b-** Write SQL commands and relational algebra expressions to retrieve the following data from the database:
- 1- The first and last names of the employees who work in Department 30.
 - 2- The first and last names of the employees who are "C Programmers".
 - 3- The last and first names of the "Java Programmers" in Department 30 listed alphabetically by last name.

EMPLOYEE			
EMPL_ID	LAST_NAME	FIRST_NAME	DEPT_ID
1001	Brown	George	20
1002	Lee	Harry	20
1003	Kirk	Bernard	30
1004	James	William	30
1005	James	Harry	40

SKILL	
SKILL_ID	SKILL_DESC
110	C Programmer
111	Systems Analyst
112	Project Manager
113	Java Programmer
114	Database Administrator

EMPLOYEE_SKILL_SET		
EMPL_ID	SKILL_ID	DATE_QUALIFIED
1001	110	9/1/1990
1001	114	7/10/1995
1002	111	5/10/1992
1002	112	6/12/1995
1003	113	6/6/2000
1004	110	9/10/1999
1004	113	6/8/2004
1005	110	1/15/1996
1005	111	2/1/1999
1005	112	3/3/2001

DEPARTMENT			
DEPT_ID	DEPT_NAME	DEPT_LOCATION	DEPT_EXTENTION
20	ERP Development	Room 101	5-2007
30	Technical Assistance	Room 103	5-2004
40	Programming Support	Room 105	5-2008